

Daily Problem: March 5, 2026

Problem Statement

Construct a Turing machine that takes as input a number n in binary and accepts if and only if n is a perfect square.

Solution

Let the input be a binary string

$$n \in \{0, 1\}^*$$

representing a non-negative integer in binary. The goal is to design a Turing machine that accepts exactly when

$$n = k^2$$

for some integer $k \geq 0$.

The construction uses the idea that we can compute successive squares and compare them with the input.

Step 1: Tape Layout

The machine uses multiple work regions on the tape.

$$\text{Input } n \mid \text{current } k \mid \text{value } k^2$$

The symbols are separated by special markers such as $\#$.

Initially the tape looks like

$$n\#0\#0$$

where

- the first block stores the input number n
- the second block stores the current counter k
- the third block stores k^2

Step 2: Iteratively Compute Squares

The machine repeatedly increases k and updates k^2 .

The key identity used is

$$(k + 1)^2 = k^2 + 2k + 1$$

The machine performs the following steps:

1. Increment k by 1 using standard binary increment.
2. Compute $2k$ by shifting the binary representation of k left by one position.
3. Add $2k + 1$ to the current value of k^2 using binary addition.
4. Compare the resulting value with the input n .

Step 3: Comparison with Input

After computing the new square value:

- If $k^2 = n$, the machine enters the accept state.
- If $k^2 > n$, the machine enters the reject state.
- Otherwise the machine continues to the next iteration.

Binary comparison is performed by aligning the most significant bits and scanning left to right.

Step 4: Correctness

At the beginning we have

$$k = 0, \quad k^2 = 0$$

After each iteration the machine updates k^2 according to

$$(k + 1)^2 = k^2 + 2k + 1$$

Thus the machine generates the sequence

$$0, 1, 4, 9, 16, 25, \dots$$

which are exactly the perfect squares.

If the input number n appears in this sequence, the machine accepts. If the sequence exceeds n , then n cannot be a perfect square and the machine rejects.

Conclusion

The Turing machine iteratively generates squares using the identity

$$(k + 1)^2 = k^2 + 2k + 1$$

and compares each square with the input number n written in binary. If a match occurs, the machine accepts; if the square exceeds n , the machine rejects. Therefore the machine accepts if and only if the input binary number represents a perfect square.